

INGB417 Facilities Design 2025 - A Practical Facilities Design Group assignment

Assessment 4 (*Report (only Design specifications)*) [no mark, only feedback] Due 25 March 2025 (this serves as the input to Design alternatives, *if there is no hand in of this report, there will be no marking of Assessment 5*)

Assessment 5 (full *Report and Interviews*) [/100] Report due on 5 May 2025 with Interviews on Tuesday 6 May 2025 (bring MFP A3 size)

This is a Group assignment. (Group size = 3) Members are assigned by the lecturer. When handing in the documents, please note the members' names and student numbers and everyone's individual contribution in a table on the cover page. The reason for a group is to provide an opportunity to learn from each other, so please don't let any individual do things completely alone without sharing what was done. This project is preparation for the GA exam.

Work must be typed. All calculations, graphs, tables, or charts must be shown to indicate insight into how the facility was designed. Provide evidence for thoughts you had, assumptions you made, the logic you used in getting to the design. The report and interview are your *evidence*.

This is a DESIGN project. The expectation is that you develop the ECSA Graduate Attribute(GA) skills specifically related to Engineering Design as applied to a Facilities Design problem.

Remember that you will have to *prove* by the way you communicate what was done, how it was done, etc. Evaluation will only be done on the written Reports and the verbal Interviews. Use words wisely. 'Sell yourselves well' For more information on how to approach this project/assignment study all 3 documents uploaded onto eFundi, as you need them:

1. INGB417 Aligning GA3 (Engineering Design process) with Facility Design Steps.pdf
2. INGB417 Clarifying Objectives Requirements and Criteria 2025.pdf
3. INGB417 Design Report guidance 2025.pdf

This group project counts 60% towards your module mark.

PURPOSE:

- To develop the skills applicable to understand a design problem, including research skills needed in *analyzing the production process design and how to calculate design requirements* (eg. Number of machines per process step).
- From this proposing and evaluating alternative solutions and *presenting a final detailed design* that solves the design problem, with appropriate justification.
- To develop *problem solving* and *creative thinking* skills
- To develop the ability and confidence to *work with people in groups* to gather appropriate and relevant data
- To develop *communication skills* in report writing and doing an interview.

THE ASSIGNMENT:

As an Industrial Engineer you have been appointed to act as the Facility Design team leader for the new facility required by Wiring.com for their new factory making 5 different types of netting, by way of 4 processes. Your task is to DESIGN a NEW MANUFACTURING FACILITY for Wiring.com. by the end of April 2025, to present to Wiring.com management on 6 May 2025.

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Refer to **Appendix 1** for the company brief you will be designing the facility for. Specifications and requirements are mentioned. If other information is needed, you are welcome to make your own assumptions based on research.

Refer to **Appendix 2** for the group members per group number. You can choose your own roles and responsibilities and who is to upload the deliverables. Only one upload per group is required.

Refer to **Appendix 3** for guidance on the type of data that you will have to get, calculate, or assume to design a facility.

DELIVERABLES:

For Assignment 4 (no mark, only detail feedback) = The **first half of your report (Design requirements)** must be PRINTED (doublesided) and handed in on 25 March 2025. The expectation is that you will by then have a clear understanding of what is supposed to be happening at the new facility and have shown enough data and analysis to start with facility layouts. This must include workstations levels showing detail dimensions as these are used to work out areas for the next steps. Feedback will be given on account of design requirements calculations.

Groups are then required to integrate findings from the data analysis to *design layouts*. Groups should present at least 3 alternative block layouts (showing calculated areas) with an appropriate evaluation process for selecting the most suitable alternative. A justification should be presented for all decisions including relevant analysis where applicable to present the chosen Final Master Facility Plan. The MFP must include scale, legend and dimensions of the plot, buildings and outer areas.

For Assignment 5 (mark /100) A **final PRINTED report (Chapters 1-10)** must be handed in (and uploaded to eFundi) on **Monday 5 May 2025 @ 8h00**. [only one upload and one printed report per group] The report must be printed and bound. The report should include the following: A cover page (reflecting the facility challenge), an Abstract (as for executive management), the Table of Contents, Lists of Figures/tables, and then the report as per the chapters indicated in the guidelines. The MFP printed on A3 can be brought to the Interview (one per group).

- A **20 minute Interview** will be done with each group on Tuesday 6 May 2025. A timetable will be set for these interviews.

ASSESSMENT of FINAL REPORT together with Interview:

Marks will be given on the following ECSA GA criteria like the exam. See Rubric to be used below.

[TIP: This is a BIG and interesting Assignment. Make sure that you start early enough to get the scope so there is enough time for later design details.

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Perform creative, procedural and non-procedural design and synthesis of components, systems, engineering works, products or processes.

EVALUATION CRITERIA				
	Weighting	Mark (Interview)	Mark (Report)	Mark (Final)
Clear interpretation of the design problem and formulation/calculation of design requirements or specifications . (includes appropriate use of analytical tools and techniques.)	/30			
Practical insight shown during the design process and in the design solution	/20			
Sufficient detail provided in the solution to meet the design requirements (Receiving & Dispatch, Processes, Storage, Offices, Support functions all in MFP)	/30			
Effective use of a structured and rigorous design process (Report structure corresponding to process steps)	/10			
Effective communication of the design process and solution	/10			
FINAL MARK (Interview 50: Report 50)	/100	Interview mark	Report total	Total mark